

The Program in Action

When you run *rot13cipher.py*, the output will look like this:

```
ROT13 Cipher, by Al Sweigart al@inventwithpython.com
```

```
Enter a message to encrypt/decrypt (or QUIT):
```

```
> Meet me by the rose bushes tonight.
```

```
The translated message is:
```

```
Zrrg zr ol gur ebfr ohfurf gbavtug.
```

```
(Copied to clipboard.)
```

```
Enter a message to encrypt/decrypt (or QUIT):
```

```
--snip--
```

How It Works

ROT13 shares a lot of code with Project 6, “Caesar Cipher,” although it’s much simpler because it always uses the key 13. Since the same code performs both the encryption and decryption (lines 27 to 39), there’s no need to ask the player which mode they want to use.

One difference is that this program maintains the casing of the original message instead of automatically converting the message to uppercase. For example, “Hello” encrypts to “Uryyb,” whereas “HELLO” encrypts to “URYYB.”

```
1. """ROT13 Cipher, by Al Sweigart al@inventwithpython.com
2. The simplest shift cipher for encrypting and decrypting text.
3. More info at https://en.wikipedia.org/wiki/ROT13
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: tiny, cryptography"""
6.
7. try:
8.     import pyperclip # pyperclip copies text to the clipboard.
9. except ImportError:
10.     pass # If pyperclip is not installed, do nothing. It's no big deal.
11.
12. # Set up the constants:
13. UPPER_LETTERS = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'
14. LOWER_LETTERS = 'abcdefghijklmnopqrstuvwxyz'
15.
16. print('ROT13 Cipher, by Al Sweigart al@inventwithpython.com')
17. print()
18.
19. while True: # Main program loop.
20.     print('Enter a message to encrypt/decrypt (or QUIT):')
21.     message = input('> ')
22.
23.     if message.upper() == 'QUIT':
24.         break # Break out of the main program loop.
25.
26.     # Rotate the letters in message by 13 characters.
```